Situation:

Let us look at the family tree of a male drone bee and a female bee as shown Fig 7.19.

Here, female bees have 2 parents, male (drone) bees have just one parent, a female. (Male bee (drone) are produced by the queen's unfertilized eggs, so male(drone) bees only have a mother but no father!)

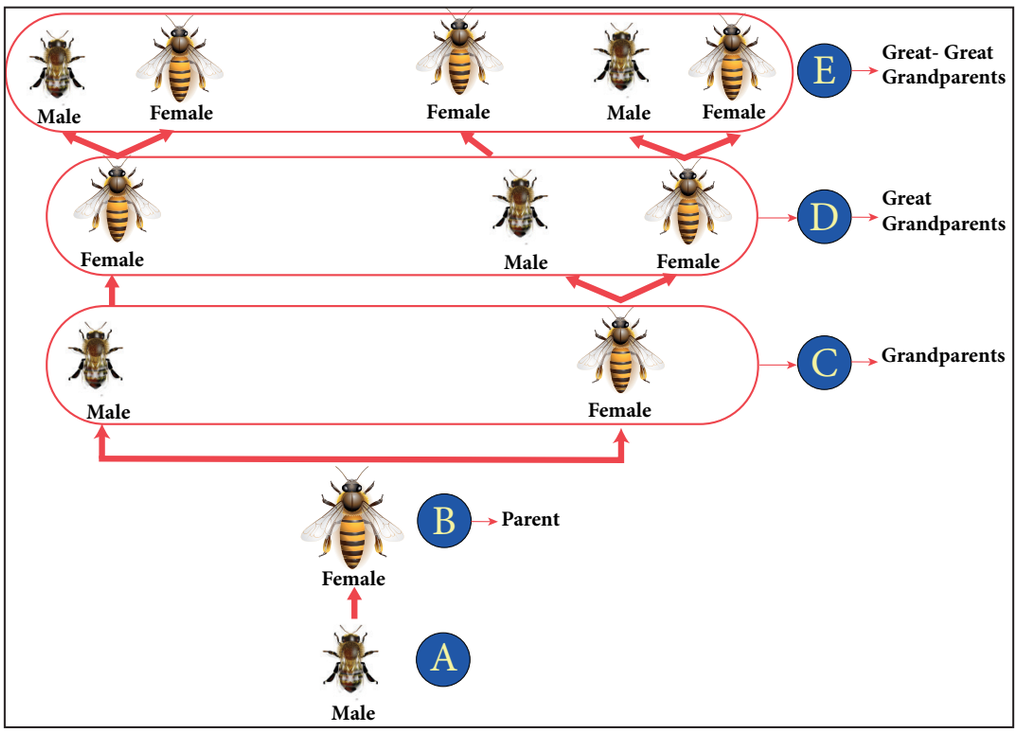


Fig. 7.19

From the picture the following points are noted:

1. The male Ⓐ has 1 parent, a female Ⓑ.
2. The male Ⓐ has **2** grandparents Ⓒ, since his mother had parents Ⓒ, a male and a female.
3. The male Ⓐ has **3** great-grandparents Ⓓ: since his grandmother has two parents but his grandfather has only one.

Now, answer, how many great-great-grandparents did the male Ⓐ have?

Let us try to find the relationship among the pattern of bees family by representing in the tabular form given below,

Number of Ⓐ	Parents Ⓑ	Grandparents Ⓒ	Great Grandparents Ⓓ	Great- Great Grandparents Ⓔ	Great- Great-Great Grandparents Ⓕ
a Male bee (1)	1	2	3	5	8
a Female bee (1)	2	3	5	8	13

We see the Fibonacci numbers 1, 1, 2, 3, 5, 8, 13… in the above table.

8th_Maths_English Medium_Chapter 7.indd 248 03-12-2022 15:22:02